IEOR 221

Homework 9

Module 10 Options – Numerical Techniques November 5, 2026

For the problems below you may implement in your preferred programming language/environment (such as Python, Excel, C++) although Python is recommended. Present both your numerical answers and your code of implementation.

This is a **team project**, to be completed with a team of 2-3 members.

1) Price a **vanilla American call option** with spot stock price S_0 = 100.0, strike K = 100.0, time to maturity T = 5 year, risk free interest rate r = 5%, continuous dividend yield q = 8%, volatility σ = 35% using a CRR binomial tree. Also calculate delta, gamma, theta, vega and rho using the techniques discussed in the lecture. Present your answers to 4 decimal places. Explain how comfortable you are about the accuracy of your numerical results.

2) Price a **European average price call option** with strike K = 100.0 and time to maturity T = 0.5 year:

$$payoff = \max\left(\frac{1}{n}\sum_{i=1}^{n} S_{t_i} - K, 0\right)$$

where the average is computed with the daily closing price, and there are 125 (= 250 × 0.5) trading days in the period (thus n = 125).

Assume the same market condition as 1) using Monte Carlo (MC) simulation with

(1) plain MC (i.e., simulation without applying any variance reduction techniques)
(2) MC with antithetic method.
Suggested sampling paths

$$S_{t+\Delta t} = S_t e^{\left(r-q-\frac{\sigma^2}{2}\right)\Delta t+\sigma\Delta W_t}, \; S'_{t+\Delta t} = S'_t e^{\left(r-q-\frac{\sigma^2}{2}\right)\Delta t-\sigma\Delta W_t}$$

(3) MC with control variate method.
Suggested control function: S itself and utilizing the fact

$$E_t\left[S_T\right] = S_t e^{(r-q)(T-t)}$$

One may also explore other possibilities for the control function.

(4) MC by combining the antithetic method and the control variate method used in the step (2) and (3).

Present both prices and error estimates. Compare your answer with the BSM formula for a vanilla call with the same strike and maturity. Discuss the pricing difference.

Let the number of time step = 125 (= n). In order to compare the effect of variance reduction one should use the same total number of simulated stock paths, e.g., 20000, throughout (1) - (4). Fix a seed for your random number generator if you can (e.g., set seed of random generator = 110526) so that your results are reproducible. Present your answers to 2 decimal places. Explain how comfortable you are about the accuracy of your numerical results.

Due: **5 PM PDT Tuesday November 24, 2026**

Note: depending on the choice of software platforms and seeds for the random number generator the MC results may vary slightly (the differences will getting smaller as the # of simulated paths increases) even with same # of simulated paths. On the other hand for a given binomial tree construction the results on a tree are unambiguous.